# Peltier Module

Category: **Electrical & Electronic Components & Kits** Type: **Short SOP** Priority: **Medium** 1 labs

**Covers these lab items:** Peltzier

## What it is used for

A Peltier module transfers heat from one side to the other when supplied with direct current, enabling small-scale heating or cooling experiments. Use the specified voltage and provide adequate heat dissipation on the hot side.

## How to use

1. Inspect the module's ceramic faces, leads, and mounting surfaces for cracks or damage.
2. Check the datasheet for rated voltage, current, polarity, and thermal limits.
3. Switch off the power supply before wiring the module.
4. Mount the module with suitable thermal contact and a heat sink on the hot side.
5. Connect the power supply with correct polarity and current capacity.
6. Apply power gradually and monitor temperature on both sides using suitable sensors.
7. Switch off power and allow the module to return to a safe temperature before handling.

## ⚠ Safety

- Do not exceed the module's rated voltage, current, or temperature limits.
- Provide adequate heat sinking to prevent overheating and damage.
- Avoid bending, crushing, or applying uneven pressure to the ceramic faces.
- Disconnect power before changing wiring or mounting.

## Care & storage

- Keep contact surfaces clean, protect the leads, and store the module in a dry, cushioned container.

*Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.*